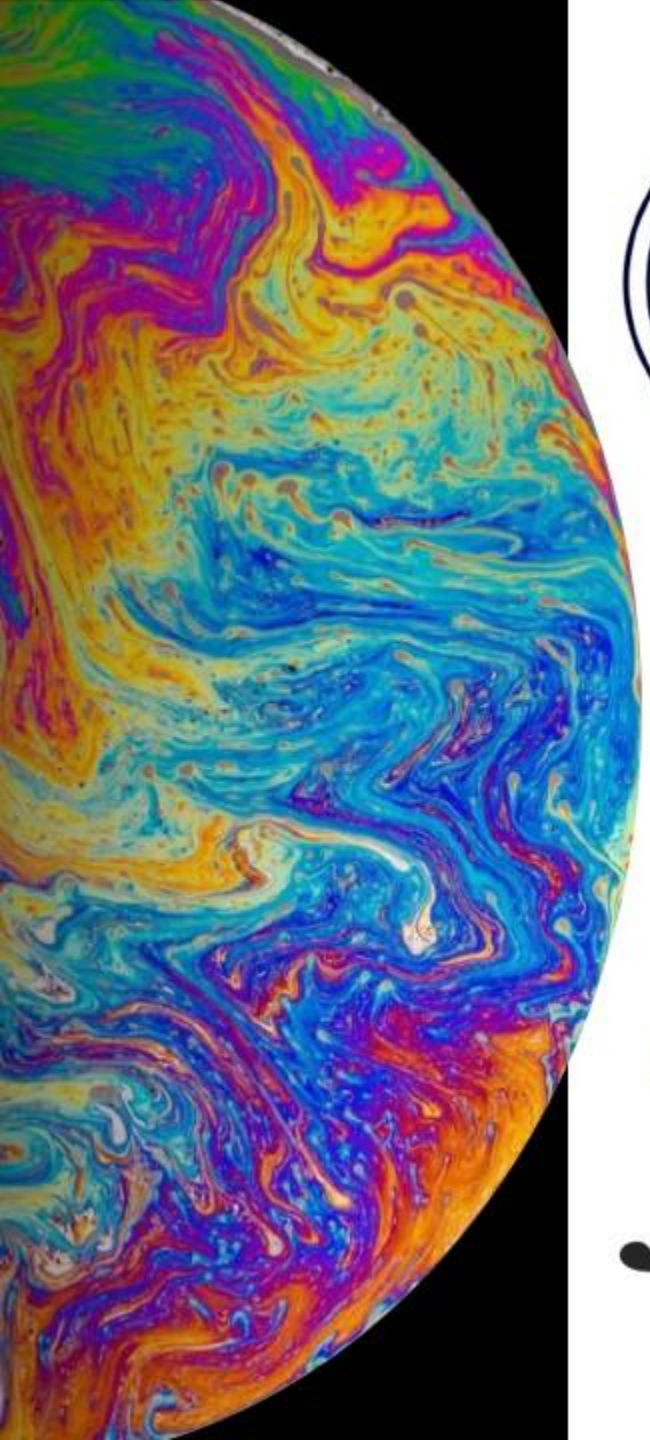


CMIP
Coupled Model Intercomparison Project

julia

# CLIMATE DATA ANALAYSIS WITH JULIA

**11. November 2026 – Online Seminar – Fresh Eyes on CMIP**




# Why Julia?

- **Readability**
  Julia is a high level programming language (like Python, R, etc.)
- **Applicability**
  Many native and open source packages for applications across many fields
  - Solving Differential Equation
  - Mathematical Syntax (supports Unicode characters)
- **Interoperability**
  R, Python and other languages are easily callable through Julia
- **Speed**
  Just in time (JIT) compilation



# JIT Compilation



## Interpreter Language

- Code is executed line by line
- Advantage: Flexible and dynamic implementation
- Problem: Slow



## Compiler Language

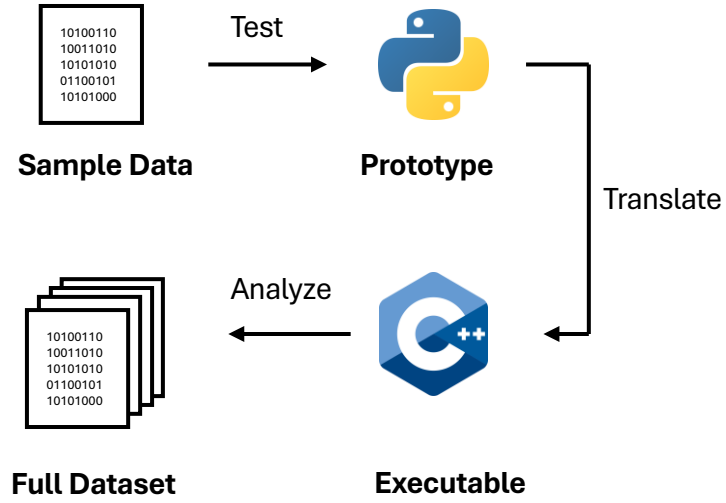
- Code is translated into machine code before execution
- Advantage: Very fast execution
- Problem: Unflexible

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## JIT – Just in Time Compilation

- Code dynamically compiled during execution
- Combining advantages of interpreter and compiler language




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**All in One**